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THE EFFECT OF EUROPEAN CULTURE ON OTTOMAN ART MUSIC

A PERSONAL EXPERIENCE OF ORAL TRADITION IN ANCIENT MUSIC

INTRODUCTION

Not from an academic background nor a professional musician, but merely an engineer and surveyor who took a deep interest in the music of an entirely different culture from which I grew up. As a person with a very inquisitive mind, I was never satisfied to accept anything at face value and felt compelled to get to the roots, source or origin of whatever subject I came into contact with. Hence when I first started to unravel the mysteries of Ottoman Art Music and the Sufi music which is associated with it, I soon realised that the use of Western-style notations was not only an imposition of a Eurocentric kind but was entirely inappropriate to the subject matter.

So from the early days of my research and practice of this music I was ever on the lookout for views and other published matters which supported this initial insight. Don't get me wrong; I was born to a traditional English Christian family in SE London just after world war two; but this entirely western upbringing did not prevent me from objectively examining the effects which European type people have had on other world cultures. These effects have come through the imposition of the (false) idea that all things European are either more advanced or are superior to those other cultures; many of which were more advanced and existed long before European culture became established.

The music of the Sufis became not only the love of my life, but a way of life for me and members of my family. So at the point where I came to the doctorate proposal, which includes writing a book on the subject, I could not leave out the chapter entitled "Misconceptions." This monograph is a preview into the basic content of that chapter¹ and related matters appearing in the doctorate dissertation called "The Music of Rumi - A Music Tradition of Pythagoras." It is all about the philosophy of music as a divine art in the way Pythagoras designed it to be.

¹ Later the book was published The Music of Rumi - A Pythagorean Music Tradition. ISBN 978 0957 1665 09

BRIEF HISTORICAL BACKGROUND TO OTTOMAN ART MUSIC

The roots of the interval structure of Ottoman Art Music and Sufi music including the concepts of their modes, share the same Pythagorean source with most music in the Middle East². This music was mainly preserved and used within the Sufi tradition. Hence its use was related to the same spiritual goal that the Sufi has³. However, it has ever been the nature of many Sufis that they live a family life and that of a householder having a role within the community in which they live. This means that not only the Sufi concepts of living but the other subjects which they find valuable, such as music, affects the life of the society in which their influences spread.⁴ Therefore it is not surprising that secular music in the Middle East has been strongly influenced by the same principles which were laid down by Pythagoras about 2,500 years ago.

The Ottomans were mixed bunch of Sultans or community leaders who gained ground and influence during the period where Islam supplanted Christianity in the area known as Byzantium. Slightly prior to this period and during the whole reign of the Ottomans in the Turkish lands, the Mevlevi dervishes developed the same concepts in music to a very high degree, and of course entirely for the benefit of the spiritual progress of the dervishes. Naturally, because of the way that the Mevlevi mingled with the Ottoman courts and the Sultans, they were in a position to influence the way in which the Ottoman court music developed⁵. In its orthodox approach Islam rejects music and other matters which Sufis find very helpful. This meant that at times when music was forbidden in society at large, it was the Mevlevi and other Sufis who became the repository of the knowledge and practice of music. Meaning that it may have died out altogether if they hadn't maintained the practice of it, albeit covertly at times.

This same concept of modes (or Makams) has become the basis of most music in the Middle East through the spread of the Ottoman influence in the lands within their vast empire. It is also the basis of the Western musical scales used in medieval times; leading on to their corrupted form as the tempered major and minor scales.

² Yekta Bey R (1921) *La Musique Turk* – section in Lavignac Encyclopaedia : Paris : Lavignac. Although the Greeks may have lost contact with Pythagorean writings, the Arab speaking peoples of the Mesopotamian area had translated into Arabic those writings in which they recognised eternal genius valuable to all peoples.

³ Sufism predates Christianity, although it may not have been called Sufism. See Brown J.P. (1868) *The Dervishes or Oriental Spiritualism* : London.

⁴ Behar C (2004) *Sufism Music and Society in Turkey and the Middle East* : London : Swedish Research Inst.

⁵ Feldman W (1996) *Music of The Ottoman Court* : Berlin : Verlag fur Wissenschaft und Bildung

THE DEMISE OF THE OTTOMANS

The Ottoman rule finally came to a close after the end of world war one when the Ottoman Empire was defeated. This was followed by the Turkish war of independence in the 1920s led by Kemal Ataturk. His subsequent agenda was to remove as much as he could which carried the influences established during Ottoman times. This included weakening the power of the Islamic Ulema. In the reforms which he introduced the Ottoman-Arabic script was banned along with aspects of the mode of dress and the banishing of all Sufi orders, including their music and confiscation of their lands and buildings. No transcriptions were commissioned of any literary works extant at the time, thus cutting subsequent generations from easy access to their history and the written works of their ancestors. This opened the way for European influences to gain the upper ground without any literal opposition. The teaching of Ottoman music was stigmatised by banning it and labelling it as corrupt, evil and backward. Yet it survived due to its innate strength in those few who made it their way of life. A truly good thing cannot become a bad thing simply because a politician chooses to call it bad. Hence, those who knew it as good endeavoured to keep it alive. It is these few that we should thank for its survival; albeit little known or practiced.

SCOPE OF THIS PAPER

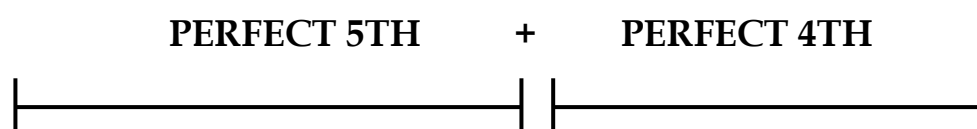
This paper will therefore look at the sources of European influence in Ottoman art music which includes, immigrant musicians, musicologists, political interference and the European teaching methods. The matters affected by these influences will discuss the construction of the modes or Makams, the number of notes in the system, the use of western style notation and the instruments played and the subsequent losses.

The influencing factors stemming from governmental interference are many, below I give the nine main points which emerged from my researches in respect of the westernisation by the new Turkish Republic.

- During the 1920s made the teaching of Ottoman music illegal.
- Sent people to study music in Europe to learn their methods.
- Commissioned theory books from those who were sent abroad.
- Invented Makams which did not exist to link the theory of the new Turkish Classical music with western music.
- Invented rhythms which did not exist for works which originally did not require a fixed rhythmic framework.
- Created music schools to replace the oral tradition (called meshk).
- Taught music from western style notations.
- Played Ottoman art music on pianos and clarinets etc.
- Replaced small music ensembles with large orchestras.

AN OVERVIEW OF THE OTTOMAN MUSIC THEORY

The Makams (musical modes) are based on ancient Pythagorean theory and are constructed of pentachords and tetrachords. These are not chords as such, i.e. they are not a group of notes played simultaneously as a chord in western style music. They are groups of notes which act as building blocks for the many melodic scales called Makams (maybe up to 200 different Makams exist, i.e. used in performance)⁶.



As shown above, the lower pitch notes are to the left side. A Makam may have a perfect fourth as the lower (left hand) section there is no hard and fast rule about this. Within the space of each perfect 4th or perfect 5th are notes formed from Just Intonation intervals. Meaning that all intervals are made from naturally occurring ratios which are easy on the human sense of listening.

Some of the Just Intonation intervals used

Name	Octave	5th	4th	large third	small third	whole tone	small whole tone	large semi tone	small semi tone
Ratio	2:1	3:2	4:3	5:4	6:5	9:8	10:9	15:14	20:19
Commas	53	31	22	17	14	9	8	5	4
Cents	1200	702	498	386	316	204	182	119	90

The sizes of the main intervals used here are shown with their equivalent sizes in 'cents' and 'commas'. The latter is the preferred way of designating intervals in the oral tradition. Commas are based on the octave being divided into 53 equal intervals. In the past it was discovered that this division of the octave gives intervals which coincide with Just Intonation intervals. Hence, these are all exact numbers of commas. Cents are the western equivalent where there are 1,200 equal intervals in the octave to suit the 12 tone tempered scale. Hence, 100 cents is one western semitone. The fact that natural intervals such as the perfect 5th and perfect 4th vary by 2 cents from an exact number of 100 cents, should not be seen as any kind of deviation from proper intervals.

⁶ Here I mean to say a Makam which has a composition associated with it. There were times during the history of the Ottomans that to have a Makam associated with your name gave prestige. Hence of the 250 or so Makams which my research has uncovered, less than 200 of these have composition so far discovered. Some of the other Makams were undoubtedly formulated for the prestige mentioned above, so they do not really exist.

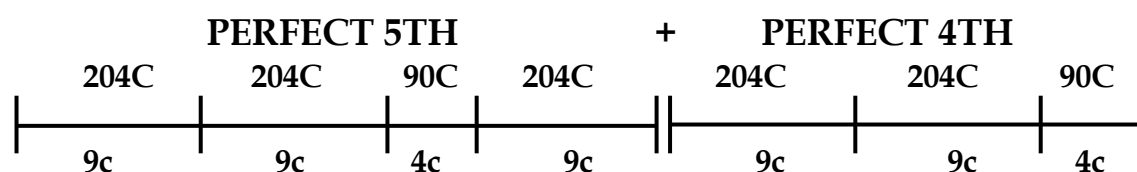
I should mention here that a Makam is much more than a musical scale. In Chapter 14 of the book 'The Music of Rumi' I describe ten basic elements which constitute a Makam. Briefly these are :-
 (1) The ambit of the pentachords and tetrachords. (2) The beginning area. (3) The final note area. (4) Dominant note or notes. (5) Temporary resting notes. (6) Ornamental notes not in the normal scale. (7) Characteristic short melodies. (8) Notes which change pitch by 4 to 5 commas. (9) Supporting tones to the tonic note. (10) The aspect of movement and rest. Altogether these create a matrix for the artist to express the deepest human emotions through music. Any single Makam is a powerful tool in a musicians hands.

There are another five special intervals used in particular Makams for specific effects. Altogether there are 11 intervals smaller than a perfect 4th. (western music has only 4). With these other five special intervals the system is Ottoman Art Music - without them it is the modern westernised idea of the latter called Turkish Art Music.

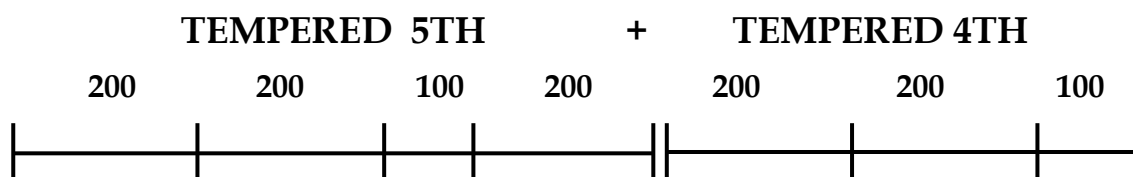
From a western musician :-

"Equal temperament is a lot easier to use than Just Intonation, but I find it lacks expressiveness. It sounds dead and lifeless to me. As soon as I began working micro-tonally, I felt like I moved from black and white into colour. I found that certain combinations of intervals moved me in a deep physical way. Everything became clear for me, more visceral and expressive."⁷

One mode called Ajemashiran was the source of the European major scale and has the intervals shown below; in cents 'C' and in commas 'c'. Commas help us to describe intervals and see the relationship between them using small numbers which we can relate to easily.

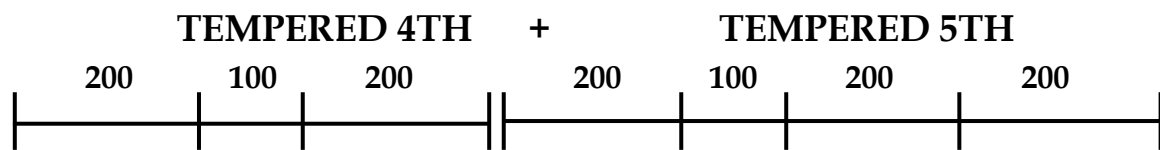


The above mode was the subject of equal temperament in Europe and produced our modern major scales. Showing cents only (i.e. commas not relevant)



⁷ Rich R. – 20th century US western music composer who uses synthesisers – Internet source.

Imitating aspects of the Makams called Puselik and Kurdi and keeping to the same groups of tempered intervals the origins of the modern minor scales can also be seen. Showing cents only.



And so for the last 150 years or more this has been the main European emphasis using tempered intervals in the teaching of musical scales.

THE NATURE OF EUROPEAN INFLUENCE

Let us try to get a sense of how the European experience of music had itself historically changed and the aspects of this which influenced Europeans when they tried to understand music of other cultures.

Music is a fleeting art which leaves the mind and emotions with an impression of some kind. It is when this impression carries a profound nature that people applaud the composer with the title 'genius'. This impression can be created with the use of both a small number of musicians and a simple use of melody. The true genius will make use of these to their best. It is when the use of both the number of musicians and numbers of simultaneous notes both become complex, that the simple impression of joy may become buried or lost altogether or replaced with some form of sensual impression. The joy I mentioned above is not sensual; it is deeper than that and more enduring.

European musical performance was once far less complex than it is in modern times. This has arisen from the endeavour to create greater impressions on the audience. The number of true geniuses in the history of European musical composers can be counted with the fingers on both hands with some left over. Yet because of their fame and the focus on the genius, many others have tried to gain the status of genius. For this they attempted to heighten the impression of their compositions through the use of more and more complex use of notes and larger groups of musicians. As if bigger or more complex is better. Over time both musicians and composers have lost contact with the simple art of creating that profound impression (joy) with the melody of music. Instead it has become the norm to create wonder and excitement within an audience, not the simple lasting expression of joy. During this process, for the purpose of convenience, musical scales were altered and the twelve tone equal interval scale became dominant. This was to suit the larger ensembles of instruments (orchestras) and mechanical instru-

ments.⁸ Over several generations this had left most people with a diminished ability to distinguish small changes in tone, coupled with the firm belief that the twelve tone equal interval tempered scale was the basis of the art of music as a whole.

The above goes some way to understanding how a modern western academic could publish the following statement.

“It has sometimes been suggested that Oriental musicians, their hearing unaffected by the artificialities of equal temperament, have preserved the capability of recognising and of applying to subtle aesthetic effect minute shades of intonation which the Western musician no longer perceived as meaningful. But this *pleasant fancy* has never been proved. It is true that two intervals 20 cents ($1/9^{\text{th}}$ of a whole tone or one comma) larger than the other, can be clearly told apart under laboratory conditions, or when attention is particularly directed to them. Nevertheless such a *fine distinction* would be as difficult for the musician to maintain throughout the course of a performance as for the audience to recognise.”⁹

Tanburi Nejdēt Yashar was tested as to his ability to hear subtle differences in tones as part of the University of Washington’s ongoing programme of discovering the comparative abilities of great performers. Below are the results from testing four specifically Ottoman intervals over a range of four or so pitch positions.¹⁰ These measurements were made by the University using a stroboconn electronic measuring device. A microphone passes the tones produced into an amplifier type device which then uses strobe lights to indicate the coincidence of the tone with a preset value. This can be varied to measure small deviations.

Washington University results

Interval cents	Interval commas	Measured (in cents) 1 comma = 22.7 cents.	Error range commas
114	5	112, 113, 117, 116, 111,	$1/8^{\text{th}}$ to $1/24^{\text{th}}$
180	8	183, 180, 176,	$1/6^{\text{th}}$ to $1/24^{\text{th}}$
272	12	271, 270, 272, 273	$1/12^{\text{th}}$ to $1/24^{\text{th}}$
144	6.33	145, 143, 143,	$1/24^{\text{th}}$

Here it is clear that the ‘*pleasant fancy*’ has been proved, albeit some years after the publication of Dr. Owen Wright’s book. Also it is not one

⁸ By the term ‘mechanical’ I mean to say such things as organs and pianos and saxophone type instruments. These instruments were a departure from the ancient idea of an instrument, where the musician’s hands, fingers or lips were intimately associated with the creation of the melodic tones. Hence machinery began to take over the note making where the melodic tones are actually made remotely from the musician’s mouth, fingers or hands. Hence the feelings within the musician could no longer affect or modify the tone produced.

⁹ Wright O – The Modal System of Arab and Persian Music – (Oxford 1978)

¹⁰ 1982 test results on Nejdēt Yashar playing his tanbur which he tuned himself to the chosen intervals.

comma difference but a small fraction of a comma (i.e. $1/24^{\text{th}}$ in some cases)¹¹. One comma to a properly trained ear is not a '**fine distinction.**' And in addition I have many recordings of Nejdēt Yashar playing his tanbur and I can verify that he has the ability to maintain the exact intervals during very rapid delivery of a stream of notes. Verified by Professor Lorraine Sakata in her comment in Nejdēt Yashar's biography.

"It was not until 1980-81 when Necdet Yaşar was again invited back to teach at the University of Washington, that I realised how his artistry was highly dependent on his deep knowledge and acute aural sensitivity of Makam intervals. A music program interested in the scientific study of music and cognition asked Necdet to take a test based on hearing minute interval differences. I was told that of all the musicians tested, both Western and non-Western musicians, he had the keenest sense of hearing and was able to discern the smallest differences in pitches. I am convinced that his acute sense of hearing and his impeccable performance standards made him the best classical Turkish musician I have had the honour of hearing in my life."¹² (note:- Lorraine calls it 'classical Turkish' not Ottoman.)

From the above we can begin to understand the basis from which western musicians and musicologists began to document their investigations into Ottoman Art Music during the 19th and 20th centuries. With the tempered twelve tone scale as their basis for understanding music theory, they needed to relate what they had read and heard to their own pre-conceived concept of what music was. Which in many cases is limited by the idea that a semi-tone ($4\frac{1}{2}$ commas) was the smallest tone required in musical scales. Whereas in Ottoman music the oral tradition requires a musician to distinguish $1/3^{\text{rd}}$ of a comma (7 or so cents).

From the above western view we can begin to understand how an American musicologist went to Turkey, mixed with many fine musicians, began to learn to play the reed flute called the Ney and after all that, his conclusion about the system of the Makams was :-

"What we have here is fundamentally a twelve tone system with two versions of each note one slightly higher than the other."¹³

And he showed a diagram using part of the tanbur neck with frets spaced about 1 comma apart :-



Karl Signell's Tanbur fret system

¹¹ For practical purposes the ability to discern $1/3^{\text{rd}}$ of a comma is sufficient in playing Ottoman Art music.

¹² Sakata Prof. L (2009) Section in - Tanburi Necdet Yaşar Biography (no single author) : Istanbul : Brainstorm

¹³ From the book called Makam – by Karl Signell.



Fret system used by normal Tanbur players

The normal tanbur neck has been used on all Turkish tanburs, even after the westernisation of the Makam system. Karl Signell conveniently omitted the additional frets shown, to support his idea that the system is a variation of the 12 tone western idea of musical scales. As shown, he also added one fret which does not exist to fill in a gap on his diagram. This view is decidedly Eurocentric of an American kind. It is false and very misleading when coming from an academic.

This is what I mean about some western people, being unable to truly see the music of other cultures. Their view is governed by the preconceived idea that the 12 tone tempered scale is the basis for all music theory.¹⁴

Hence, due to the semi-tone being their smallest interval, at one time westerners erroneously concluded that intervals smaller than one semi-tone were unnecessary for making music. Later in 1932 in a conference in Cairo this was modified to $\frac{1}{4}$ of a whole tone to suit D'Erlanger's 19th century study and theory of Arabic music. Since then mainstream Arabic music has been westernised and has lost its original splendour.

Baron D'Erlanger appeared to list a large number of Ottoman Makams in his book "La Musique Arabe". Many of these Makams were actually invented by Ottoman composers. This was during the times when the Ottoman Empire spread its influence throughout the Arab speaking world. In his book he uses quarter tones to describe every interval in all the 120 Makams listed.

This concept of his is looking at Middle Eastern music as if it were a variation of the 12 tone equally tempered scale instead of the other way around. Once more a Eurocentric view.

His notation system uses 5 accidentals and only 6 intervals smaller than the 4th. If we translate quarter tones into commas

LA MUSIQUE ARABE;
SES RÈGLES ET LEUR HISTOIRE

55. LE MODE 'UŠŠĀQ-TURKĪ

55— Echelle du Mode 'UŠŠĀQ TURKĪ

FIG. 100.

EXTRACT FROM LA MUSIQUE ARABE

¹⁴ Nejdett Yashar's tanburs had several other frets not shown here. These were to suit the need for expressing the Makams with the full gamut of intervals in accordance with tradition. In his home he had more than a dozen tanburs, some of which had even more frets so that he could play where a transposed situation or a special Makam requirement had a need for other frets. He never compromised the need for correct expression.

every interval is sufficiently incorrect to alter the sound of the music significantly.

This idea results in a loss of 8 intervals, and many more notes in the system by merging notes close to each other as a single note. This idea was clearly a deciding influence in the choices finally settled on by the Turkish musicians of Ataturk's regime, appointed to westernise the Turkish music theory. This they did as if the system were the special property of Turkey and they had the right to do that.¹⁵

So now let's look at notation and its effect.

THE EFFECT OF USING MUSICAL NOTATION

Professor Paulos, who has researched the use of notations in Ottoman music very thoroughly, presented some of his findings at a seminar in London University on 9th April 2009, which I attended. He has provided me with a written quote of his answer to a question on whether notations were used for performance by Ottoman musicians :-

"There is no evidence suggesting the wide use of notation systems in the performance of Ottoman classical music, certainly up until the mid 19th century. Until the systematic adoption of the Western notation system (beginning of the 20th C.) we encounter a number of diverse notation systems (alphabetic-numerical, neumatic [Greek Orthodox Church], Hamparsum etc.) which are used either in the context of musical collections (Ali Ufki etc.) or in that of theoretical treatises (Cantemir).¹⁶"

Hence the previous use of notation was not for performance but for record or theoretical purposes. Despite this history, the Turkish musicians who were commissioned to westernise Turkish Classical music (previously Ottoman Art music) decided that all known works should be notated in a consistent format, so that music schools could teach people to play from notation. For this they chose to use the western five line staff system. This however presented them with a number of difficulties. The five line staff designates notes which are of a certain pitch and distinct from each other. In Ottoman music performance adjacent notes often blend into each other by sliding techniques. There is no adequate way to indicate this on the five line staff. Similarly notes may waver up or down to suit the Makam situation during a performance. This cannot be indicated in notation either, without apparently fixing a thing which is not fixed. Also with the single flat and single

¹⁵ Most of the great Turkish teachers who taught me various aspects of the music were unequivocal in their opinion that, the traditional Ottoman music was a subject so deep, that it really belonged to the whole of humanity.

¹⁶ Professor Pangiotis Paulos – Athens University – Greece.

sharp system used in the West they needed to introduce more flats and sharps so as to include the notes needed to play all the Makams.

Yet there were many in Turkey who opposed the introduction of western style notation for Ottoman music; both for teaching it and for performance. This debate has gone on for many decades since those early days (now nearly a century). Ruhi Ayangil in his long article about Turkish notation systems in the Journal of the Royal Asiatic Society, concludes that in the 21st century there still is debate in Turkey as to how many notes make the music system and whether or not the system can be notated using a Western style with the five lines :-

“Time after time, --- there have been those who claimed that it would be better to rely solely on the oral tradition system (meshk) without the use of any notation.”¹⁷

However, back in the 1930s Ezgi and Arel studied the attempts at notation made by previous musicians in the history of Ottoman music. Two such musicians were western immigrants to the Ottoman Empire. Ali Ufki (Boboski) was a Polish musician who gained favour in the Ottoman courts he used the five line staff system in his notations. The other, Cantemir, was a Romanian prince who had been trained in western music methods. He designated a list of notes with Arabic type letters or characters.

Eventually the notation method adopted was called the Ezgi Arel system. Which was a collaboration between Dr. Suphi Ezgi and Sadettin Arel. This system ignored the system of Rauf Yekta Bey, a Mevlevi musician and musicologist; and the system of Ekrem Karadeniz, who had a greater depth of understanding of the Makam system. The Ezgi Arel system designated 24 notes in an octave using four different sharp signs and four different flat signs to indicate the notes. Below is a table of the various accidentals (flats and sharps) used by the above mentioned musicians and theorists. Cms = commas.

Rauf Yekta Bey				Ezgi Arel				Ekrem Karadeniz			
Flats		Sharps		Flats		Sharps		Flats		Sharps	
Sign	Cms	Sign	Cms	Sign	Cms	Sign	Cms	Sign	Cms	Sign	Cms
↓	1	#	1	↓	1	#	1	↓	1	#	1.5
↓	3.33	#	4	↓	4	#	4	↓	2	#	3
↓	4	#	5	↓	5	#	5	↓	3.5	#	4
↓	5	#	8	↓	8	#	8	↓	5	#	5.5

¹⁷ Ayangil R – (Cambridge 2008).

All systems recognise the need for the 5 and 4 comma intervals; (15:14 and 20:19), but this is insufficient to preserve the magic of the music.

Rauf Yekta Bey was like many previous Ottoman musicians who delved into the use of notation of the Makams. He presumed that any person who was in a position to play an Ottoman instrument would have been through aspects of teaching from a master of the Makam system (meshk)¹⁸. Hence the musician would have been in a position to know that certain notes would require changes of fractions of a comma for their actual performance where the Makam requires this. So his system only shows one accidental to designate a part comma. In other words he saw no need to make an attempt to designate on notation all the notes required to be played. We should not interpret his system as a statement of the total number of notes in the Makam system.

Ekrem Karadeniz, on the other hand, foresaw the future situation where musicians would be trying to pick up knowledge of the Makams mainly from notation, or from a teacher who had not been taught all the subtleties of the Makams. Hence his system attempts to allow notation to instruct the performer as close as possible, the actual tones to be played. This is the reason for the fractions of commas.

The Ezgi Arel system has flats and sharps which mirror each other, in the same way the European accidentals do. Meaning that the same notes can be derived either using flat of the note above or sharp of the note below. It would appear that Ezgi and Arel had used a similar judgement which Owen Wright later adopted. One of assuming that small differences in tone were both unnecessary and could not be distinguished by the human ear; neither performer nor audience.

So we see at least three ideas of notation, but I have given a little of the thinking behind them which is important to have clear when assessing the basis for their choices. Many other systems of notation have been tried before and since the Ezgi Arel ideas. These are discussed in the book which formed my doctorate presentation - now published and called "The Music of Rumi."

I should also note here that prior to the idea that notes will be called 'do sharp' or 're flat' etc. the meshk system attributed names to all the notes required for all the Makams. Each note had a character with a name, which in many cases was associated with a Makam of that same name. Teachers and musicians would have taught or discussed the compositions and the Makam constructions using

¹⁸ The section on Political Decisions where meshk is described.

these names. So everyone knew what they were talking about without any need for notation methods. This use of individual note names gave them a mental map of the music system.

The main influence which has arisen from these ideas of using notation arise from the false concept that Makam performance is possible in the way western music is performed. Meaning that notation predetermines the tones and the actual numbers of notes to play. Whereas the traditional Ottoman performance would not seek to emulate or reproduce any previous performance. Musicians would interpret the original composition individually and to suit the instruments in the ensemble, the venue and the audience. Performers would be applauded for the originality of their individual interpretation. This way of performing composed works is taboo in western music recitals. The original composers scripts must be adhered to. Most classical Ottoman works did not originally have a script from a composer they were handed down through the oral system of instruction. Scripts appeared later from notating performances. Contrary to some ethnomusicologist's views, this does not mean that the original compositions were altered or corrupted over time. This would again be applying a western concept of rigidity. The Ottoman performance system gave each musician the opportunity to apply his/her own creativity to the overall plan of the composition, thereby bringing freshness to the music. This concept is close to the western idea of jazz.

Let me also contribute here another facet of Ottoman performance not much discussed elsewhere. Compositions were not generally created for any specific instrument to play their melodies. Thus any combination of instruments or indeed any single instrument could perform a composition, because they all consist of a single melodic line. This also means that the pitch basis for the performance was not fixed either. The various instruments used in Ottoman music all had differing natural pitch levels which suited their own timbre. Over time the range of pitch levels to be used was embodied in the range of Neys (reed flutes) this range has an octave span from the Bolahenk Ney to the Nisfiye Bolahenk Ney.¹⁹ So when several different instruments decide to play together they decide on which of the Ney pitches to play. This may entail some of the instruments playing on a pitch transposed from their normal pitch. This transposition would be a perfect 4th or perfect 5th in most cases.

So notation has had the above effects along with the effect of reducing the number of tones and the actual frequency of many tones in the

¹⁹ See my paper 'The Middle Eastern Ney' where most aspects of the Ney and its construction are discussed.

modern performance of Ottoman compositions, in what is now known as a Turkish Classical music concert.

THE EFFECT OF POLITICAL DECISIONS

There is no doubt that many nails in the coffin of Ottoman art music were driven in during, and subsequent to, the huge reforms made to every aspect of Turkish society by the regime set up by Kemal Ataturk. These reforms were basically aimed at completing the process of the Europeanisation of the Turkish nation, which had commenced during the closing decades of the Ottoman Empire. The reforms were enacted with all embracing laws rigidly enforced.

Ataturk was educated partly in Europe and became convinced that European ways were the best formula for the “modernisation” of Turkey. Hence he used the power behind the idea that Europe’s values were a desirable means of escape from what he deemed, the backward, corrupt and evil ways of the Ottomans. To support his purposes he passed laws which made writing with Arabic script illegal. It was very quickly replaced by the use of the Romanised alphabet similar to that used by Europeans. In time this prevented Turkish people from accessing information which may have helped in the confusion of how to play the music correctly.

Creating a notation system which was both inadequate and incorrect, did not change the way the older musicians played their traditional instruments without the need for notation. Teaching music Ottoman style may have been illegal at one time, but playing it was not. Hence conflicts were inevitable between the new Europeanised theory and practice in relation to the true understanding of what a Makam was. In addition to this, those who were appointed to oversee the new system of musical education in Turkey were pro-European. They did not know the value of the traditional system called ‘meshk’, which entailed a suitable student approaching a master of a particular instrument to teach both the art of the music and the art of the instrument. Only those who were devoted to the music would be taught through the meshk system. Payment of fees was not considered good practice.

Meshk was powerful and effective in transmitting the entire music system to the next generation of musicians. It was basically a combination of the powers of memory and imitation. It required a person with exceptional abilities in these skills. In the 1980s I sat with Necdet Yashar for many hours during many visits over several years, while he

taught me more than 70 Makams (all tape recorded with his consent). He usually played two or three different examples of good compositions in each Makam, along with excellent improvisations in these Makams. He never once referred to a notation and each piece was played perfectly. Do the maths; this means he held in his memory around 250 to 300 compositions, at least. This was the power of meshk. His knowledge of each Makam was impeccable and often at variance to the way these appeared in the various theory books extant at that time. He was able to fully describe the errors in the theory books for me so that I would not later become confused. Those lessons were meshk. Nejdet was emphatic that the system had been corrupted by the modern ideas in theory books and in the teaching methods prevalent at that time.

It was Nejdet, during his visit to my home town in Surrey UK, who asked me and my family to start an organisation in the UK where we were politically free to do so, so that the music system could be preserved as he had taught it. As an example of how restricted he was at that time - when we decided to call the charity, "The Ottoman Art Music Society of Great Britain." he was worryingly alarmed that he would be associated with anything which carried the name 'Ottoman'. "I could get into big trouble." he said. So we changed the name from Ottoman to Turkish. Today the same worries would not be there.

The political decision to side-step the meshk system and replace it with a European music academy approach was big nail in the coffin of Ottoman Art Music, it was a failing in the decisions made at that time.

LOSS OF INTERVAL STRUCTURE

The main end product of the above westernisation of the traditional Ottoman and Mevlevi use of the Pythagorean modes, has lead to a loss in the available intervals which modern musicians have, according to the theory as taught in modern Turkey. In the book 'The Music of Rumi' I describe the intervals used as coming from two ways of applying the Just Intonation series. The simple Just Intonation set arises from the series 2:1, 3:2, 4:3, 5:4, etc. The second series arises from combining adjacent values from the first series. So we have $4:3 \times 5:4 = 5:3$ etc. This second series gives rise to what I have called 'causal intervals'. All these intervals were verified to me during lessons with Nejdet Yashar. However, he did not use the mathematical ratios in his description, he always spoke in terms of commas. In many cases he would say, "This interval is 2, 2 ½ or 3 commas (flat or sharp)." Thus he was describing the fractions

of commas which show in the table below. The full set of intervals and the losses due to the Ezgi Arel system are portrayed in the table.

Just Intonation					Causal				
Ratio	Exact Commas	No. used	Cents	LOST	Ratio	Exact Commas	No. used	Cents	LOST
2:1	53	53	1200		3:1	84	84	1902	
3:2	31	31	702		5:3	39.06	39	884.4	
4:3	22	22	498		7:5	25.72	26	582.5	YES
5:4	17.06	17	386.3		9:7	19.22	19 ⅓	435.1	YES
6:5	13.94	14	315.6		11:9	15.34	15 ⅓	347.4	YES
7:6	11.79	12	266.9	YES	13:11	12.77	13	289.2	
9:8	9.01	9	203.9		15:13	10.94	11	247.7	YES
10:9	8.06	8	182.4		23:21	6.96	7	157.5	YES
15:14	5.28	5	119.4		25:23	6.38	6⅓	144.4	YES
16:15	4.93	5	111.7		27:25	5.88	6	133.2	YES
19:18	4.13	4	93.6		31:29	5.10	5	115.5	
20:19	3.92	4	88.8		39:37	4.02	4	91.14	
77:76	1.0	1	22.64						

Table of Intervals used in the Pythagorean Melodic System²⁰

Here we see that at least eight intervals are lost and this translates to many more notes in the system. The losses can be seen to be mainly on the causal interval side. It is the causal intervals which provide an essential element of the magic and charm of the original music when it is played traditionally. The Westernisation has given the music a taste which is more acceptable to the European / western ear, which has been educated to accept the tempered (artificial) set of intervals. This education has now become the norm in Turkey and has therefore deprived the nation of the ability to know the difference. Except that many old recordings exists of musicians who were trained to play the correct interval structure. Hence the debate about what to do with Turkish music theory is kept alive even in this 21st century.

CONCLUDING REMARKS

It is not too late for the Turkish nation to recover the charm and originality of their rightful heritage in music. Modern Turkey is embracing the concept that can stand on its own feet and does not need the influence of Europe and the West as much as it did after the second world war. It is fast becoming a strong nation with many aspects displaying originality. Yet recently (2017) I have seen a Turkish music recital where traditional instruments were playing old Ottoman songs, along with the accompaniment of clarinets and pianos. I am amazed

²⁰ How these intervals are used to form musical scales is fully described in Chapter 12.

that the performers and the audiences of such recitals, cannot see both how ridiculous such an ensemble appears, nor can they hear how horrible the music is, which arises from tempered instruments combined with those that are attempting to play the correct Ezgi Arel intervals. It would seem that the mind of the Turkish nation has been corrupted as well as the music. So that they now cannot see or hear the obvious.

If the Turkish nation could fully wake up to the corrupting influence which the westernisation of their traditional music has had, then they would bring about reforms to eliminate all aspects of that influence. There are many recordings and videos still available which show both visually and aurally how their ancient music was performed. So they have no real excuse to carry on in the way that they do. It requires a simple determination to keep the performance of western music and Ottoman (Traditional Turkish) music in entirely different camps.

The full knowledge may be contained in many written works about Ottoman music. The system as taught by Nejdēt Yashar is fully recorded in my doctorate work - "The Music of Rumi" - of which there are many copies in Turkey.

I rest my case.

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